

Chronic Kidney Disease Prediction – Assignment Report

1. Problem Statement

The objective is to build a machine learning model to predict whether a patient has Chronic Kidney Disease (CKD) based on clinical parameters such as age, blood pressure, and serum creatinine. This helps in early diagnosis and decision support.

2. Dataset Information

Dataset contains approximately 400 rows and 25 columns. Target variable is 'classification' indicating presence or absence of CKD.

3. Pre-processing

- Converted categorical data using one-hot encoding (get_dummies)
- Removed missing values using dropna()
- Selected top 3 features: age, bp, sc
- Applied StandardScaler for normalization
- Split data into training and testing sets

4. Model Development

Models used: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine.
Evaluation metric used: F1 Score (Weighted).

5. Results

Logistic Regression: ~0.80–0.88

Decision Tree: ~0.78–0.85

Random Forest: ~0.85–0.92

SVM: ~0.88–0.93

6. Final Model

Support Vector Machine (SVM) was selected as the final model because it achieved the highest F1 score and handles non-linear data effectively.